TNPCB CERTIFIED ECO-NETWORK • CIRCULAR ECONOMY PLATFORM

Eco Vision Tamil Nadu

Smart Waste Management, Collector GPS Navigation & Citizen Payout Platform

Platform: Full-Stack Web Application

Stack: React 18, Vite, Fastify, Leaflet, Tailwind, MongoDB

Coverage: All 38 Districts of Tamil Nadu

Date: October 2026

1. Executive Summary & Purpose

EcoVision is an intelligent environmental resource recovery and route optimization platform tailored for the state of **Tamil Nadu**. The platform bridges the gap between waste generators (citizens, residential apartments, IT corridors, and MSME industries) and authorized waste recovery facilities certified by the **Tamil Nadu Pollution Control Board (TNPCB)**.

By combining precision geospatial navigation (OpenStreetMap, Leaflet, and Haversine proximity formulas), real-time scrap market valuations, carbon offset analytics, and automated doorstep pickup dispatching, EcoVision incentivizes circular economy practices and diverts recyclable materials from local landfills.

📍 Precision GPS Routing

Pinpoint detection, draggable origin markers, and instant turn-by-turn routing via Google Maps.

💰 Scrap Monetization

Real-time pricing for plastics, e-waste, metals, paper, and textiles with instant UPI/Cash estimates.

🌿 Carbon Accounting

Scientific calculation of metric CO_2 emissions saved and equivalent trees planted per booking.

2. Complete Technology Stack

Layer	Technologies	Role & Architecture Benefit
Frontend Core	React 18.3, Vite 6.1, ES Modules	Ultra-fast hot module replacement, optimized production bundles, modular component architecture.
Styling & UI	Tailwind CSS 3.4, Material-UI 6.4	Glassmorphism dark/light design system, accessible dialogs, alerts, and smooth transitions.
Mapping & GIS	Leaflet 1.9, React-Leaflet 4.2	Custom interactive map, draggable user origin pin, dynamic navigation polylines, uncertainty radius.
Geocoding	OpenStreetMap Nominatim API	Reverse geocoding (lat/long to street address) and Tamil Nadu locality/pincode search autocomplete.
Backend API	Fastify 5.x, Node.js v24	High-throughput Node.js microframework (up to 30,000 req/sec), JSON schema validation, CORS.
Data Layer	Mongoose 8, MongoDB + Memory Fallback	Primary persistent storage in MongoDB with auto-switching in-memory seed store for resilient uptime.

3. System Architecture & Component Hierarchy



4. Detailed End-to-End Workflows

Workflow 1: User Geolocation & Precision Origin Pinpointing

1 **GPS Detection with High Accuracy & Cache Busting**

The citizen clicks "Use My GPS". `App.jsx` invokes `navigator.geolocation.getCurrentPosition` with `enableHighAccuracy: true` and `maximumAge: 0`. Captures the device's coordinates and accuracy radius in meters.

2 **Automatic Reverse Geocoding via Nominatim**

Coordinates are dispatched to `geocoding.js`. Nominatim API resolves the coordinates into human-readable localities (e.g. *"Anna Nagar East, Chennai"* or *"Gandhipuram, Coimbatore"*) with memory caching.

3 Fine-Tuning: Draggable Pin & Address Search

Because Windows PCs/laptops lack satellite GPS hardware and estimate location via ISP IP, users can drag the green origin pin directly to their house. Alternatively, `LocationSearchBar.jsx` allows instant typing of any Tamil Nadu street or 6-digit pincode.

Workflow 2: Category Filtering & Haversine Distance Engine

1 Waste Category Selection

User selects a category (e.g., E-Waste, Plastics, Metals). `loadCollectors()` triggers with parameters `{ wasteType, lat, lng, district }`.

2 Mathematical Distance Calculation (Haversine Formula)

In `collectorRoutes.js`, distance in km is calculated between user coordinates and each registered hub:
`distance = 2 * 6371 * atan2(sqrt(a), sqrt(1 - a))`
City drive duration is calculated at an average urban speed of 35 km/h: `durationMin = max(5, round((distanceKm / 35) * 60))`.

Workflow 3: Dynamic Mapping & Turn-by-Turn Navigation

1. **Color-Coded Marker Mapping**
   Collectors display with distinct color-coded SVG pins according to category (Cyan for Plastics, Purple for E-waste, Amber for Metals, Blue for Paper, Emerald for Organic). Distance badges float on each pin.

2. **Active Route Polyline & Guidance Panel**
   Selecting a collector draws an animated dashed route line on the map. `NavigationRoutePanel.jsx` activates, showing distance in km, drive time, and an **"Open in Google Maps GPS"** button launching turn-by-turn driving directions.

Workflow 4: The Citizen Booking & Monetization Pipeline

1. **Pickup Mode Selection & Address Autofill**
   User chooses between **Self Drop-off** or **Doorstep Pickup**. For doorstep pickup, a 1-click button (*"Use Map Pin"*) autofills their reverse-geocoded address directly into the form.

2. **Live Payout & Environmental Carbon Valuation**
   As the user selects or types estimated weight (5kg, 10kg, 25kg, 50kg, 100kg), the UI dynamically computes:
   Estimated Citizen Payout: `Weight (kg) × Material Market Rate (₹/kg)`
   Carbon Offset: `Weight (kg) × Material $CO_2$ Emission Factor (kg $CO_2$)`

3. **Order Dispatch & Eco-Pass Generation**
   Submitting the form issues a `POST /api/bookings` request. Fastify generates a certified pass (`ECO-TN-XXXXXX`), persists the record, and the UI triggers a confetti celebration with the confirmed pass.

5. Waste Material Categories & Valuation Matrix

Waste Stream	Accepted Subtypes	Avg. Payout Rate	Carbon Offset	Environmental Impact
Plastics & Polymers	PET bottles, HDPE containers, LDPE film, PP sacks	₹20 / kg	1.8 kg CO_2 / kg	Recycled into polyester fiber, pellets, and recycled resin.
E-Waste & Electronics	Smartphones, motherboards, PCBs, laptop batteries, chargers	₹60 / kg	14.5 kg CO_2 / kg	Precious metal recovery (Gold, Copper, Palladium); toxic lead diversion.
Metals & Scrap Alloys	Aluminum cans, copper wiring, brass fittings, iron scrap	₹40 / kg	4.2 kg CO_2 / kg	Smelted into structural rebar, engine castings, and ingots.
Paper & Cardboard	Corrugated carton boxes, office A4, newspaper sheaves	₹15 / kg	1.2 kg CO_2 / kg	Repulped into packaging liners, tissue, and paperboard.
Textiles & Garment Offcuts	Tiruppur knitwear cuttings, denim, discarded cotton shirts	₹24 / kg	3.5 kg CO_2 / kg	Shredded into shoddy yarn, thermal insulation, and wiping rags.
Glass & Cullet	Beverage bottles, glass jars, broken architectural panes	₹8 / kg	0.8 kg CO_2 / kg	Crushed into furnace cullet, reducing kiln temperatures by 30%.
Organic & Bio-waste	Household kitchen wet waste, hotel leftover food, horticultural green	₹5 / kg	0.9 kg CO_2 / kg	Converted into bio-methane gas and nutrient-rich agricultural compost.
Hazardous & Biomedical	Used motor oil, lead-acid batteries, discarded clinical kits	Free Safe Disposal	8.0 kg CO_2 / kg	Authorized high-temperature incineration and chemical stabilization.

6. Complete REST API Specifications

Method	Endpoint	Query / Body Params	Response Details
GET	`/api/health`	None	Returns server uptime, database mode, and timestamp.
GET	`/api/collectors`	`wasteType`, `district`, `lat`, `lng`, `search`	List of matching centers with computed distanceKm and durationMin.
GET	`/api/collectors/:id`	URL Param `id`	Complete center profile: verified registration, materials, capacity, hours.
POST	`/api/collectors/nearest`	`{ lat, lng, wasteType, maxDistanceKm }`	Top 5 closest centers strictly ranked by proximity coordinates.
GET	`/api/districts`	None	All 38 districts with registered facility counts and coordinates.
GET	`/api/categories`	None	All 8 waste streams, market rates per kg, and CO2 offset multipliers.
POST	`/api/bookings`	`{ collectorId, type, wasteCategory, weight, name, phone, address, date }`	Creates booking, calculates payout & carbon, returns ECO-TN-XXXXXX pass.
GET	`/api/bookings`	None	Returns recent citizen collection and drop-off records.
GET	`/api/stats`	None	Aggregated state metrics: total kg recycled, CO2 saved, certified centers.

7. Data Schema Architecture

Collector Schema (Mongoose)

```
{
  name: String,
  city: String,
  district: String,
  address: String,
  lat: Number,
  lng: Number,
  phone: String,
  acceptedWasteTypes: [{
    category: String,
    ratePerKg: Number
  }],
  operatingHours: String,
  capacityKgPerDay: Number,
  isVerified: Boolean,
  tnpcbRegNumber: String,
  rating: Number
}
```

Booking Schema (Mongoose)

```
{
  bookingId: String, // ECO-TN-XXXXXX
  collectorId: String,
  collectorName: String,
  type: String, // dropoff | doorstep_pickup
  wasteCategory: String,
  estimatedWeightKg: Number,
  userName: String,
  userPhone: String,
  userAddress: String,
  district: String,
  scheduledDate: String,
  timeSlot: String,
  estimatedReward: Number,
  co2SavedKg: Number,
  status: String // confirmed | completed
}
```

8. Installation & Execution Guide

To run the entire EcoVision platform locally on your machine:

```
# 1. Install all dependencies across root, server, and client:
npm run install:all

# 2. Start both the Fastify Backend (Port 5000) and React Client (Port 3000):
npm run dev

# 3. Individual microservice commands:
npm run server  # Fastify API (http://localhost:5000)
npm run client  # Vite Frontend (http://localhost:3000)
```